LESSON 116 (2.0) GROUND

Lesson Objective

During this lesson, the applicant will become familiar with the regulations applicable to an applicant pilot preparing for solo flight including both 14 CFR part 61 and 91. This will include a detailed discussion on applicant pilot privileges and limitations, endorsements required for solo flight, airworthiness requirements and pilot qualifications. The learner will also be introduced to CFIT, wire strike avoidance, SRM, and aviation security

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
    - Quiz the student on the relevant sections of the Flight Operations Manual:
- Student questions.

References

- Pilot's Handbook of Aeronautical Knowledge. (FAA-H-8083-25B)
    - Chapter 2: Aeronautical Decision Making
    - Chapter 9: Flight Manuals and other documents
    - Chapter 15: Airspace
- Advisory Circular. (Search Tool)
    - Parts 91 and 135 Single Pilot, Flight School Procedures During Taxi Operations. (AC 91-73)
    - Certification of Pilots and Flight and Ground Instructors. (AC 61-65)
- Federal Aviation Regulations
- Aeronautical Information Manual (AIM).
    - Chapter 07 - Safety of Flight.
        - Section 04 - Wake Turbulence.
        - Section 06 - Potential Flight Hazards.
- FTA Flight Operations Manual.
    - Part 02 - Technical.
        - Section 02 - Ramp Operations.
        - Section 03 - Flight Operations
- Controlled Flight into terrain (CFIT)

Instructor Guidance

- This lesson is a good opportunity to complete ground instruction when a flight is cancelled. If weather or aircraft availability causes a flight lesson to cancel, use this lesson to make productive use of the time.
- Due to the large volume of material, covering 2–3 topics per session and marking the lesson incomplete is acceptable. Do not rush through the content — depth of understanding matters far more than checking every box in a single sitting.
- Ensure students are provided with the recommended resources so they can effectively review these topics independently between sessions.
- This lesson ensures students have the required knowledge at the **student pilot level** to safely solo the aircraft from the airport, through Class D airspace, to the practice area, and back. Keep that context front and center — everything covered today is directly tied to their ability to operate safely and legally as a solo pilot.
- These topics will be revisited in Stage 2 and Stage 3 with greater depth. Avoid going beyond what is operationally relevant for a student pilot at this stage — the goal is a solid foundation, not exhaustive regulatory mastery.
- Applicants are not required to memorize SRM acronyms, but they should understand their purpose and be able to apply them in context. Focus on the decision-making process behind the models rather than rote memorization of letters.
- When covering endorsements required for solo flight, make sure the student knows exactly where those endorsements must be located — in their logbook. Walk them through what each endorsement looks like and explain what happens if one is missing or expired before a solo flight.
- When discussing personal minimums, have the student actually write theirs down before leaving the lesson. Completion standard number six requires that they have written personal limitations — do not let this be a verbal exercise only.
- The go/no-go decision discussion should be interactive. Ask the student to walk through a scenario using PAVE and IMSAFE rather than simply explaining the models to them. Students who apply these tools in a scenario during the lesson will use them more naturally in the aircraft.
- Ensure the student has Florida Tech Security and Melbourne Airport Police Department contact numbers before leaving this lesson, as referenced in the aviation security section.
- Since the pre-solo tests in Lesson 121 cover regulations and FOM content directly, use this lesson as an opportunity to flag areas where the student is weak and assign targeted review accordingly.

Lesson Content

Regulations Applicable to an Applicant Pilot

- Ensure that the student has the FARAIM opened and follows along.
- Before going through the regulations, highlight and explain the different parts and subparts of the FAR.
    - Part 1: definitions and abbreviations.
    - Part 23: airworthiness standards for normal category airplanes.
    - Part 43: Maintenance, preventive maintenance, rebuilding, and alteration.
    - Part 61: Certification: Pilots, flight instructors, and ground instructors.
        - Subpart A: General
        - Subpart B: Ratings
        - Subpart C: Student Pilot
        - Subpart D: Recreational Pilot
        - Subpart E: Private Pilot
        - Subpart F: Commercial Pilot
        - Subpart G: ATP
        - Subpart H: CFI
    - Part 67: Medical standards and certification.
    - Part 71: Designation of class A, B, C, D, and E airspace areas; air traffic service routes; and reporting points.
    - Part 73: Special use airspace.
    - Part 91: General operating and flight rules.
    - Part 141: Pilot schools.
        - Differentiate between part 61 and part 141.
- Part 61
    - Required documents - 61.3:
        - Pilot certificate, medical, and government photo identification when operating as a required pilot.
    - Drugs and alcohol - 61.15 and 61.16:
        - Outline the consequences.
    - Student pilot certificate duration - 61.19:
        - No expiration date if issued after April 1st 2016.
    - Medical certificate classes and durations - 61.23:
        - Special issuance, SODA, etc. (67.401).
    - Type rating of aircraft 61.31
        - Aircraft need a type rating if:
            - Large Aircraft (more than 12,500 lbs)
            - Turbojet-Powered Aircraft
            - Any aircraft deemed necessary by the administrator
    - Pilot Logbooks - 61.51:
        - Information to be logged.
    - Medical Deficiency - 61.53:

        - What constitutes a medical deficiency.
    - Flight Review - 61.56:
        - Every 24 calendar months.
        - At least an hour ground and an hour flight that must cover the legally required areas.
        - Must receive endorsement to complete.
    - Recent flight experience - 61.57:
        - Acting as PIC while carrying passengers during day time vs night time.
        - Same category, class, and type.
        - If tailwheel, needs to be to a full stop.
        - At night to a full stop.
    - Change of address - 61.60:
        - The FAA must be notified in writing within 30 days.
    - Solo requirements - 61.87:
        - This will be covered under endorsements required for solo flight below.
    - General limitations (SP) - 61.89:
        - This will be covered under student pilot limitations FAA and FIT below.
    - Solo operations in class b airspace - 61.95:
        - This will be covered under endorsements required for solo flight below.
- Part 91
    - Pilot in Command (PIC) responsibilities - 91.3:
        - PIC is responsible for operation of the aircraft, safety of flight, and has the final authority.
        - This includes determining airworthiness.
    - Aircraft airworthiness - 91.7:
        - Cannot fly unless airworthy.
    - Flight manual, placards, and markings - 91.9:
        - All must be on the airplane to be airworthy.
    - Careless and reckless operation - 91.13:
        - May not operate in a manner that endangers others, on the ground or in the air.
    - Dropping objects - 91.15:
        - Can drop objects if the right precautions are taken.
    - Drugs and alcohol - 91.17:
        - Cannot be under the influence.
        - 8 hours “bottle to throttle” and a maximum BAC of 0.04%.
        - Compared to FIT’s regulation.
        - Must submit to a drug/alcohol test if requested.
        - Must furnish results to the FAA within 24 hours.
    - Preflight action - 91.103:
        - IFR flight or away from the vicinity of an airport.
            - **N**OTAMs.
            - **W**eather.
            - **K**nown ATC delays.

        - **R**unway lengths.
        - **A**lternates if needed.
        - **F**uel requirements.
        - **T**akeoff/landing data.
  - Crewmembers at stations
    - 91.105:
    - When must required crew members be at their stations?
    - When should seatbelts and shoulder harnesses be worn?
  - Use of safety belts and harnesses
    - 91.107:
    - All occupants must be briefed on how to use seatbelts and shoulder harnesses.
    - When should seatbelts and shoulder harnesses be worn?
  - Operating near other aircraft
    - 91.111:
    - May not operate an aircraft so close to another that it creates a collision hazard.
    - Review formation flight.
  - Right of way
    - 91.113:
    - See and avoid other aircraft if the weather permits!
    - Review all of the different scenarios and who has the right of way.
  - Speed limits
    - 91.117:
    - 250 kts below 10,000 ft MSL (Applies in Bravo).
    - 200 kts below Class B or in VFR Corridor.
    - 200 kts within 4 nm and 2,500’ Class D or C primary airport.
    - May use minimum safe speed if needed.
  - Minimum safe altitudes
    - 91.119
    - Congested
      - 1,000 ft above, 2,000 ft horizontally.
    - Other than congested
      - 500 ft above.
    - Sparsely populated or open water
      - cannot operate within 500' of the nearest vessel, structure, person, or vehicle.
  - Altimeter Settings
    - 91.121
  - Compliance with ATC
    - 91.123:
    - Must obey instructions unless amended clearance or emergency.
    - Must notify ATC immediately when deviating.
    - May need to submit a report to ATC within 48 hours.

- Temporary Flight Restrictions (TFRs) - 91.137-145:
  - These are where TFR NOTAMs derive authority.
- FAA fuel reserves
  - 91.151:
  - You must have fuel to fly from departure to destination, destination to any alternates, and then fly thereafter at cruise power setting for:
    - Day: 30 mins.
    - Night: 45 mins.
    - FIT: 60 mins day or night.
- VFR cruising altitudes
  - 91.159:
  - Purpose is for separation on converging courses.
  - Based off of the magnetic course starting at or above 3,000 ft
    - 360-179 degrees → odd 1000's + 500.
    - 180 - 359 degrees → even 1000's + 500.
- Required aircraft equipment
  - 91.205
  - This will be further explained in the airworthiness section.
- Emergency Locator Transmitters
  - 91.207
  - When are they needed?
  - How often must they be inspected?
  - What are they inspected for?
  - How often do you need to change the battery?
- Aircraft lights
  - 91.209:
  - Used while the aircraft is being operated.

- May be turned off when they are unsafe.

Airworthiness Requirements

- Define airworthiness.
  - Type Certificate Data Sheet.
    - Ensure the student knows where to find it.
      - Faa.gov/aircraft → technical information → TCDS → search your aircraft.
  - Supplemental Type Certificate.
  - Inspections (AVIATES):
    - **A**nnual.
    - **V**OR.
    - **1**00 hours.
    - **A**ltimeter.
    - **T**ransponder.
    - **E**LT.
    - **S**tatic system.

- Airworthiness directives:
  - Define what an airworthiness directive is.
  - Three types:
    - Notice of proposed rulemaking.
    - Final rule.
    - Emergency.
- Service bulletins:
  - Define what a service bulletin is and how it is different from an AD.

- Required documents and equipment for VFR flight.
  - Applicable regulations: 91.203 and 91.205.
  - Required documents (ARROWES):
    - **A**irworthiness certificate.
      - Does not expire but invalid if not airworthy.
      - Check tail number and signature during pre-flight.
    - **R**egistration certificate.
      - Typically expires every 7 years.
      - Check tail number and expiration date during pre-flight.
      - Different ways to lose the registration certificate:
        - Death.
        - Transfer.
        - Aircraft is found doing illegal activities.
        - Aircraft is totaled.
        - Certificate holder requests in writing to cancel their registration.
        - Certificate holder loses US Citizenship.
    - **R**adio station license / radio operator permit.
      - Obtained through the Federal Communications Commission (FCC).
      - Radio station license is for the airplane.
        - Expires every 10 years.
      - Radio operator permit is for the pilot.
        - Does not expire.
    - **O**perational limitations.
      - POH section 02.
    - **W**eight and balance.
      - POH section 06.
    - **E**quipment list.
      - POH section 02.
      - Difference between MEL and KOEL.
    - **S**upplements:
      - POH section 09.
  - Required equipment:
    - Day VFR required instruments (ATOMATOFLAMES) - 91.205.
      - **A**irspeed Indicator.
      - **T**achometer.

            - For each engine.
        - **O**il Pressure.
        - **M**anifold Pressure.
            - For each altitude engine (turbocharged/supercharged).
        - **A**ltimeter.
        - **T**emperature Gauge.
            - For each liquid cooled engine.
        - **O**il temperature.
        - **F**uel Quantity Gauge.
        - **L**anding Gear Position Lights.
            - If retractable.
        - **A**nti-Collision Lights.
            - Aircraft certified after March 11th, 1996.
        - **M**agnetic direction indicator.
        - **E**LT.
        - **S**eatbelts.
    - Night VFR required instruments (FLAPS) - 91.205.
        - **F**uses/resettable circuit breakers.
        - **L**anding Light (Electric) for hire.
        - **A**nti-Collision Lights.
            - Aircraft certified after August 11th, 1971.
            - Aircraft certified before that date STILL NEED anti-collision lights, but the standards/requirements are different.
        - **P**osition Lights.
        - **S**ource of Electricity (Alternator/Battery).
- Inoperative equipment - 91.213:
    - If an aircraft has a Minimum Equipment List, then refer to that first and find out if the aircraft is still airworthy.
    - Determining Airworthiness with inoperative equipment without MEL (TARK):
        - **T**ype certificate data sheet (TCDS).
        - **A**irworthiness Directives.
        - **R**equired for 91.205.
        - **K**OEL, Kinds of Operating Equipment List (section 2 of POH).
        - If the instrument or equipment is NOT required:
            - Remove from the airplane in accordance to 43.9 or deactivate and placard inoperative.
- Preventive Maintenance
    - Part 43, Appendix A, Paragraph (c)

Pilot Qualifications

- Medical certificates - 61.23.
    - This was reviewed under part 61 regulations above.
- Student pilot certification - 61.83.

    - Eligibility requirements - 61.83.
    - Application - 61.85.
    - Required endorsements were reviewed above.
- Private pilot certification - 61.102.
    - Eligibility requirements - 61.103.
    - Aeronautical experience - 61.109
        - Highlight the differences between part 61 and part 141.
    - Endorsements required for check ride/end-of-course.
        - Aeronautical knowledge test - 61.35(a), 61.103(d), and 61.105.
        - Flight proficiency - 61.103(f), 61.107(b), and 61.109.
        - Prerequisite for practical test - 61.39(a)(6)(iii).
    - Privileges and limitations - 61.113.
- Basic Med
    - Eligibility
        - Hold a valid FAA pilot certificate (student pilots cannot use BasicMed).
        - Hold a valid U.S. driver's license.
        - Have held a valid FAA medical certificate at some point after July 14, 2006.
        - Have not had their most recent medical certificate denied, suspended, or revoked
    - Maintain Eligibility
        - Physical Examination (Every 48 Months)
            - Must be completed with a state-licensed physician.
            - Uses FAA Form 8700-2 (Comprehensive Medical Examination Checklist – CMEC).
            - The physician confirms that no known medical condition interferes with safe aircraft operation.
        - Online Medical Education Course (Every 24 Months)
            - Pilots must complete an FAA-approved BasicMed course such as: AOPA BasicMed Course, Mayo Clinic BasicMed Course
    - Limitations
        - Aircraft maximum certificated takeoff weight: 12,500 lbs or less.
        - Aircraft no more than 7 seats and 6 passengers maximum.
        - Maximum altitude below 18,000 ft MSL.
        - Maximum speed 250 knots or less.
        - No operations for compensation or hire.
        - Flights generally within the United States unless authorized by another country.
- Require logbook entries - 61.51.

National Airspace System (BASIC INTRODUCTION – THE REST WILL BE INTRODUCED IN STAGE 2)

- Controlled airspace (91.127 - 91.135).
    - General term that refers to the different classifications of airspace within which air traffic control (ATC) service is provided. (STAGE 1 WILL ONLY FOCUS ON THE FOLLOWING):
        - Class D:

                - Generally, from SFC to 2,500 ft MSL.
                - Surrounds airports that have a control tower.
                - Configuration can be different depending on the airport (KMLB!).
                - Usually designed to contain instrument approaches.
                - Entry requirements:
                    - Establish two-way radio communication (ATC MUST say your call-sign).
                - Required equipment:
                    - Two-way radios.
            - Class E:
                - Controlled airspace that is not class A, B, C, or D.
                - Extends up from either the surface or a designated altitude to the overlying or adjacent controlled airspace.
                - Sectional and other charts depict all locations of Class E airspace with bases below 14,500 feet MSL. In areas where charts do not depict a class E base, class E begins at 14,500 feet MSL.
                - In most areas, the Class E airspace base is 1,200 feet AGL. In many other areas, the Class E airspace base is either the surface or 700 feet AGL.
                - All airspace above FL 600 is Class E airspace.
                - Required equipment:
                    - ADSB-OUT: at or above 10,000 feet msl, excluding airspace at and below 2,500 feet agl; over the Gulf of Mexico, at and above 3,000 feet msl, within 12 nm of the U.S. coast.
                    - Mode C transponder: at and above 10,000 feet msl, excluding the airspace at and below 2,500 ft agl.
- Uncontrolled airspace (91.126)
    - Portion of the airspace system that is not designated as A, B, C, D, or E.
        - Class G:
            - Uncontrolled.
            - ATC has no authority or responsibility.
                - Although ATC has no authority, a pilot cannot be careless or reckless as this would break the 91.13 regulation.
            - VFR rules apply.

- VFR weather minimums:

Basic VFR Weather Minimums

Airspace			Flight Visibility	Distance from Clouds
Class A			Not applicable	Not applicable
Class B			3 statute miles	Clear of clouds
Class C			3 statute miles	1,000 feet above 500 feet below 2,000 feet horizontal
Class D			3 statute miles	1,000 feet above 500 feet below 2,000 feet horizontal
Class E	At or above 10,000 feet MSL		5 statute miles	1,000 feet above 1,000 feet below 1 statute mile horizontal
	Less than 10,000 feet MSL		3 statute miles	1,000 feet above 500 feet below 2,000 feet horizontal
Class G	1,200 feet or less above the surface (regardless of MSL altitude).	Day, except as provided in section 91.155(b)	1 statute mile	Clear of clouds
		Night, except as provided in section 91.155(b)	3 statute miles	1,000 feet above 500 feet below 2,000 feet horizontal
	More than 1,200 feet above the surface but less than 10,000 feet MSL.	Day	1 statute mile	1,000 feet above 500 feet below 2,000 feet horizontal
		Night	3 statute miles	1,000 feet above 500 feet below 2,000 feet horizontal
	More than 1,200 feet above the surface and at or above 10,000 feet MSL.		5 statute miles	1,000 feet above 1,000 feet below 1 statute mile horizontal

Figure 15-8. *Visual flight rule weather minimums.*

  - Special use airspace:
    - Where certain activities must be confined.
    - Limitations may be imposed on operations not participating.
    - Information for airspaces is located on aeronautical charts.
    - ONLY FOCUS ON THE FOLLOWING FOR STAGE 1:
      - Restricted area:
        - Area in which operations are hazardous to non-participating aircraft.
          - Artillery firing, guided missiles.
        - Flight of aircraft may be restricted, but not prohibited.
        - Must have permission from ATC or the controlling agency to enter.
        - Charted with an "R" followed by a number.
        - Information found on the back of charts.
      - Military Operation Area (MOA):
        - Defined vertical and lateral limits.
        - Separates certain military training activities from IFR traffic.
        - IFR traffic may be cleared through airspace if ATC can provide separation.
        - Additional information for each found on the back of the sectional.
    - Other airspace (ONLY FOCUS ON THE FOLLOWING):
      - Parachute jump aircraft operations:
        - Airports at which there are high numbers of parachute operations.
        - Published in chart supplement under Parachute Jumping Areas and depicted on sectional charts.
      - Temporary flight restrictions (TFRs):
        - Designated by flight data center (FDC) Notice to Airmen (NOTAM):

        - Begins with phrase "FLIGHT RESTRICTIONS" followed by location, effective time period, and defined in statute miles, and altitudes affected.
    - Purposes may include:
        - Protect persons or property in air or on surface from existing or imminent hazard.
        - Provide a safe environment for operation of disaster relief aircraft.
        - Prevent unsafe congestion of sightseeing aircraft above incident or event.
        - Protect declared national disasters for humanitarian reasons in Hawaii.
        - Protect the President, Vice President, or other public figures.
        - Provide a safe environment for space agency operations.
    - If penetrated, the pilot may undergo security investigations and certificate suspensions.

Applicant Pilot Limitations

- Review student pilot limitations as outlined in the Florida Tech Aviation FOM.
- During dual instructional flights, operating in conditions outside typical student pilot solo limitations can be beneficial when conducted safely.
- As the primary instructor, use your judgment based on the student's skill level and proficiency.
- Weather conditions can deteriorate unexpectedly during solo flights; providing the student with supervised experience in more challenging conditions helps prepare them for real-world situations.

Endorsements Required for Solo Flight

- Pre-solo aeronautical knowledge: 14 CFR § 61.87(b)
- Pre-solo flight training: 14 CFR § 61.87(c)(1) and (2)
- Initial solo flight endorsement (first 90 calendar days): 14 CFR § 61.87(n)
- Solo flight endorsement renewal (each additional 90 calendar days): 14 CFR § 61.87(p)
- Solo takeoffs and landings at another airport within 25 NM: 14 CFR § 61.93(b)(1)
- Additional endorsements will be required later in training, including solo cross-country endorsements and checkride endorsements prior to practical test submission.

Personal Minimums

- Personal minimums are self-imposed operating limits that pilots establish to maintain a safe margin beyond regulatory requirements. These limits are based on a pilot's experience level, proficiency, and comfort level.
- Personal minimums help pilots make safe go/no-go decisions by setting clear limits for conditions such as:
    - Weather (visibility, ceilings, winds)
    - Crosswind component
    - Runway length and airport conditions

    - Pilot fatigue and stress levels
    - Aircraft performance and equipment status
    - Establishing and adhering to personal minimums helps reduce risk, improve decision-making, and promote safer flight operations, especially as conditions approach a pilot's limits.
- PAVE Model
    - The PAVE Model is a risk management tool used by pilots to identify hazards and evaluate risks before and during a flight.
    - The four areas evaluated are:
        - P – Pilot: The pilot's physical and mental condition, experience, and proficiency.
        - A – Aircraft: The aircraft's airworthiness, equipment, and performance capabilities.
        - V – enVironment: External conditions such as weather, terrain, airspace, and airport conditions.
        - E – External Pressures: Outside influences that may affect decision-making, such as time constraints or passenger expectations.
    - The PAVE Model helps pilots assess potential risks and make safe go/no-go decisions during flight planning and operations.

- IMSAFE Checklist
    - The IMSAFE checklist is a personal safety tool used by pilots to evaluate their physical and mental condition before flight.
        - **I** – Illness: Any sickness that could affect performance.
        - **M** – Medication: Over-the-counter or prescription drugs that may impair judgment or coordination.
        - **S** – Stress: Personal or professional stress that could distract from flying.
        - **A** – Alcohol: Compliance with the "8 hours bottle to throttle" rule and no impairment.
        - **F** – Fatigue: Lack of adequate rest that could affect alertness and performance.
        - **E** – Emotion / Eating: Emotional state or lack of proper nutrition that may affect decision-making.
    - Using the IMSAFE checklist helps pilots determine whether they are physically and mentally fit to safely operate an aircraft.

Go or No-Go Decision

- A **go/no-go decision** is the process a pilot uses to determine whether a flight should be conducted, delayed, or cancelled based on current conditions and risk factors.
- Pilots evaluate factors such as:
    - Weather conditions and forecasts
    - Aircraft airworthiness and performance
    - Pilot readiness and proficiency
    - Airport and airspace conditions
    - External pressures

- This decision should be made during preflight planning and continuously evaluated throughout the flight. Using risk management tools such as **PAVE, IMSAFE, and personal minimums** helps pilots make safe and informed go/no-go decisions.

Wake Turbulence Avoidance

- Explain how wake turbulence is created:
  - Pressure differential → low pressure above and high pressure below the wing.
  - Air moves from the high-pressure area under the wing to the low-pressure area over the wing.
    - It goes around the wing tip and then trails behind.
    - As air curls around the tip, it combines with the wash to form a fast spinning trailing vortex.
    - Right wing tip - counterclockwise.
    - Left wing tip - clockwise.
- Explain how to avoid wake turbulence:
  - Vortices are greatest when an aircraft is in a heavy, slow, and clean configuration.
    - Avoid flying through another aircraft's flightpath.
    - Rotate prior to proceeding aircraft.
    - Delay takeoff by 2-3 minutes.
    - Avoid following aircraft on similar flight paths.
    - Land past the touchdown point of the other aircraft.
  - Wind is an important factor to wake turbulence.

Runway Incursion Avoidance

- Define a runway incursion.
- Highlight the 4 categories (INTRODUCE):
  - Category A: serious incident in which a collision was narrowly avoided.
  - Category B: incident in which separation decreases and there is a significant potential for collision.
  - Category C: incident characterized by ample time and/or distance to avoid a collision.
  - Category D: incident that meets the definition of runway incursion such as incorrect presence of a single vehicle/person/aircraft on the protected area of a surface designated for the landing and takeoff of aircraft but with no immediate safety consequences.
- Explain how to avoid a runway incursion:
  - Always brief taxi instructions.
  - Be familiar with the airport layout.
  - Know where hold-short lines are located.
  - Do not cross a hold-short line without obtaining clearance.
- Ensure the runway is clear of vehicle/person/aircraft before taking-off and landing.

Controlled Flight into Terrain (CFIT)

- Controlled Flight Into Terrain (CFIT) occurs when an airworthy aircraft, under pilot control, is unintentionally flown into terrain, water, or an obstacle without the pilot being aware of the impending collision.
- CFIT accidents often result from a loss of situational awareness due to factors such as poor visibility, night operations, weather conditions, or unfamiliar terrain.
- To avoid CFIT, pilots should:
    - Maintain proper situational awareness of terrain and obstacles
    - Follow minimum safe altitudes
    - Use available navigation equipment and charts
    - Monitor weather and visibility conditions
    - Avoid descending below safe altitudes without proper visual references.

Wire Strike Avoidance

- Wire strikes occur when an aircraft collides with power lines, cables, or other wires, often during low-altitude operations. These obstacles can be difficult to see, especially in poor lighting or against certain backgrounds.
- To avoid wire strikes, pilots should:
    - Avoid unnecessary low-level flight.
    - Maintain adequate altitude when operating near obstacles.
    - Be especially cautious near roads, rivers, valleys, and towers, where wires are commonly present.
    - Identify supporting structures such as poles or towers, which are often easier to see than the wires themselves.
    - Maintain strong situational awareness during takeoff, landing, and low-altitude maneuvering.
- Proper planning, obstacle awareness, and maintaining safe altitudes are the best defenses against wire strike accidents.
- Review AIM 7-6-4: Obstructions to Flight

Single Pilot Crew Resource Management (SRM)

- Single-Pilot Resource Management (SRM) is about how to gather information, analyze it, and make decisions. Learning how to identify problems, analyze the information, and make informed and timely decisions is not as straightforward as the training involved in learning specific maneuvers
- Aeronautical Decision Making (ADM)
    - A systematic approach to decision making used by pilots to determine the best course of action in response to a given situation.
- DECIDE Model
    - The DECIDE Model is a structured decision-making process used by pilots to safely respond to changing conditions during flight.
    - The steps are:
        - **D** – Detect a change that requires attention.

- **E** – Estimate the need to react to the change.
- **C** – Choose the best course of action.
- **I** – Identify the actions needed to carry out the decision.
- **D** – Do the necessary action.
- **E** – Evaluate the results of the action.
- The DECIDE Model helps pilots make logical and effective decisions when faced with unexpected situations or hazards.

- 5P Model
  - The 5P Model is a risk management tool used to help pilots evaluate key factors that affect the safety of a flight. It encourages pilots to continuously assess conditions before and during flight to identify potential risks and make safe decisions.
  - The five areas evaluated are:
    - **Pilot** – Physical and mental condition (IMSAFE).
    - **Plane** – Aircraft airworthiness, fuel, and performance.
    - **Plan** – Weather, route, airspace, and overall flight planning.
    - **Passengers** – Passenger expectations, distractions, and safety.
    - **Programming** – Management of avionics, GPS, and automation.
  - The 5P Model should be reviewed regularly throughout the flight to maintain situational awareness and support sound decision making.
- CARE Model
  - The CARE Model is a risk management tool used to help pilots identify hazards and determine the level of risk before and during flight operations.
  - The four steps are:
    - **C** – Consequences: Identify the possible outcomes if a hazard occurs.
    - **A** – Alternatives: Consider different courses of action to reduce risk.
    - **R** – Reality: Evaluate the likelihood that the hazard will occur.
    - **E** – External Pressures: Recognize outside influences that may affect decision-making, such as time constraints or passenger expectations.
  - The CARE Model helps pilots make safer decisions by encouraging thoughtful evaluation of potential risks and available options.
- **T**EAM Model
  - The TEAM Model is a risk management tool used to help pilots assess and manage risks by considering available resources and possible mitigation strategies.
- The four steps are:

    - **T** – Transfer: Shift the risk to another resource when possible (e.g., request ATC assistance or delay the flight).
    - **E** – Eliminate: Remove the hazard completely (e.g., avoid hazardous weather or cancel the flight).
    - **A** – Accept: Accept the risk if it is minimal and manageable.
    - **M** – Mitigate: Reduce the level of risk by taking corrective actions (e.g., choosing an alternate route or delaying departure).

- The TEAM Model helps pilots determine the best strategy for managing risks identified during flight planning and operations.

Aviation Security

- Aviation security focuses on preventing unlawful interference with aircraft, passengers, and airport operations. Pilots must remain vigilant and aware of potential security threats both on the ground and in flight.
- Pilots should:
  - Secure the aircraft when unattended.
  - Be aware of suspicious persons or activities around the aircraft or airport.
  - Follow airport security procedures and access restrictions.
  - Verify passenger identity and control access to the aircraft when appropriate.
  - Report suspicious activity to airport authorities or law enforcement.
- Maintaining awareness and following proper security practices helps ensure the safety of pilots, passengers, and airport operations.
- Make sure your student has the phone number for Florida Tech Security and Melbourne Airport Police Department.

Visual Scanning and Collision Avoidance

- Quiz the student on how to scan and avoid traffic.
- Quiz the student on right of way rules.
- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Completion Standards

This lesson is complete when the applicant can:

1. Explain regulations applicable to applicant pilots.
2. Understand airworthiness requirements and how to determine if a flight can be completed with inoperative equipment.
3. Explain the qualifications to be a applicant pilot.
4. Understand operations within the national airspace system and the rules that must be complied with.
5. List the endorsements required for solo flights and where they should be located.
6. Understand the importance of, and have written, personal limitations.
7. Be able to identify all aspects of making a competent go or no-go decision.
8. Explain the location, effects, and avoidance procedures for wake turbulence.
9. Describe a runway incursion and various mitigation techniques.
10. Understand CFIT, wire strike avoidance, SRM, and aviation security.
11. Describe proper techniques for visual scanning and its importance during all phases of flight.